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1 Abstract

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2 The problems with industrial agriculture

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Although the rise of the modern industrial structure of agricultural production has given many societies the ability to forestall famine and thrive in ways long thought impossible, many of its social and ecological repercussions have revealed themselves to be potentially dangerous in the best scenario and significant contributors to an existential crisis in the worst. These problems range from the extensive use of land, to the expulsion of pollutants and greenhouse gasses into the environment, to even social problems like the dependency of a whole population on a few producers. [Give a better overview] The following sections will explore some of these problems.

3.1 Land Use

The current paradigm of agricultural production requires the use of vast swathes of land, along with a considerable amount of resources, which inevitably ends up homogenizing the local flora and destroying the habitat of the local fauna, leading to an overall loss of biodiversity. The agricultural expansion that the current system entails is one of the main drivers of global biodiversity loss (IPBES, 2019, p. 412), and is the most widespread form of land-use change, with cropping and animal husbandry covering over one third of the terrestrial land surface (IPBES, 2019, p. XVI).

This geographical expansion of the agricultural sector has been a constant for decades, having grown by 78 million ha (780,000 km^2) between 2001 and 2023, mostly driven by an increase in permanent crops such as cocoa, oil palm or coffee (FAOSTAT, 2025). All of these are used intensively for monoculture, as are feedstock crops which by themselves account for one third of the total crops produced globally, this in turn drives massive biodiversity and habitat loss (Battersby et al., 2024, p. 22). Monoculture is another significant problem that will be explored in a subsequent section.

According the the FAO's State of Food and Agriculture report (Food and Agriculture Organization of the United Nations, 2025, p. 4), agriculture currently drives 90 percent of global deforestation, with cropland expansion accounting for almost half of the total deforested area. [Find source to quantify deforestation impact on climate change]

4 Background

Past attempts, Netherlands case study

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